PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Undifferentiated-predominant mixed-type early gastric cancer is more aggressive than pure undifferentiated-type: a systematic review and meta-analysis
AUTHORS	Yang, Peng; Zheng, Xiang-Dong; Wang, Jun-Mei; Geng, Wen-Bin; wang, xiaoyong

VERSION 1 – REVIEW

REVIEWER	Ohashi, Manabu Cancer Institute Hospital, Japanese Foundation for Cancer Research, Department of Gastroenterological Surgery
REVIEW RETURNED	02-Aug-2021

GENERAL COMMENTS	The reviewer really appreciates having an opportunity to evaluate this manuscript entitled “Undifferentiated-predominant mixed-type early gastric cancer is more aggressive than pure undifferentiated-type: a meta-analysis” written by Yang Peng et al. The manuscript is generally interesting. But the reviewer has major and minor comments listed below. Major comments 1. The authors conducted a meta-analysis regarding lymph node metastasis, SM invasion, lympho-vascular invasion, and gross type compared between MU and PU. To the best of the reviewer’s knowledge, the majority of the literature included in this study is single center, retrospective studies. In addition, due to the small number of references, the level of evidence obtained from this study seems to be low. 2. The information extracted from the literature is insufficient. Reference 18 describes depth, LVI, and ulcer findings. Reference 21 describes the depth (all cases SM) and the gross type. Reference 27 describes the depth (M in all cases). Reference 28 describes the gross type and ulcer findings. However, the authors did not indicate these data in Table 1 and did not include them in the analyses. Therefore, the authors seem to have arbitrarily selected the references for analysis. The information in each literature should be extracted again. 3. The clinical benefits of this meta-analysis are unclear. Although this study was conducted on surgical cases, the clinical merits should be clearly shown, such as the reduction of lymph node dissection in PU or the necessity of D2 dissection in MU. Minor comments 1. The introduction is redundant. If the authors would like to discuss surgical cases, the description regarding ESD is too long. The following part should be deleted. ESD has become an important method of treatment for EGC patients with a lower risk of

	LNM…which consists of both differentiated and undifferentiated histological type according to the histology. 2. In the Methods section, why is the gross type included in the comparison? The authors should provide rationale that the gross type is involved in lymph node metastasis. In addition, the reason for dividing the patients into 0-IIc and others is unclear. The authors should clearly state why they divided the patients in such fashions. 3. In the Results section, why did the authors analyze the literature from Japan or China as sub-analyses should be presented. 4. In the Discussion section which is redundant, the following sentence is also found in the Introduction section. The reviewer recommends that the authors delete it. ESD has been gaining popularity as the primary treatment for patients with EGC. However…. provide clear guidance for the clinical management of MU and PU EGC. 5. The authors also describes “Another study reported that the protein expression levels of Ki-67, extracellular matrix metalloproteinase inducer (EMMPRIN) and vascular endothelial growth factor (VEGF), which participate in angiogenesis and cell proliferation, were reduced in intestinal-type GC compared to the intestinal portion of mixed-type GC [32]”. However, the mixed-type here means the gastro-intestinal type in mucus traits. It has nothing to do with this study. The author should delete the description. 6. The conclusion is too abstract. The authors should clearly state what the results mean in clinical practice.

REVIEWER	Razmi, Mahdieh Iran University of Medical Sciences
REVIEW RETURNED	06-Oct-2021

GENERAL COMMENTS	In the present manuscript, Peng Yang et al. have compared the biological behavior of undifferentiated-predominant mixed-type (MU) EGC and pure undifferentiated-type (PU) EGC. This study identified that patients with MU EGC had an increased risk of submucosal invasion, lymphovascular invasion, depressed-type (IIc) endoscopic growth patterns and LNM compared with PU EGC patients. Manuscript is well written which will provide clarity to the scientific reader community. I am recommending this manuscript for publication after revision. 1. In the subgroup analysis, I suggest to be performed based on all countries not just for one country. 2. Data and figure regarding the ulcer findings were not presented. Authors should provide the results of this part. 3. I strongly suggest analysis based on survival outcomes to compare the mortality rate between MU EGC and PU EGC.

REVIEWER	Pripp, Are Hugo

	Oslo universitetssykehus Ulleval, Oslo Centre for Biostatistics & Epidemiology
REVIEW RETURNED	04-Nov-2021

GENERAL COMMENTS	Statistical review: Thank you for an interesting study. It should be of interest, but I have a few comments about the statistical analysis and presentation of the results. * Please also use funnelplots to assess publication bias. You may present some of them in the manuscript. * I suggest to add an example of the stata codes used for the statistical analysis. It makes the reproduction of your analysis easier. * Table 1: I suggest to add some general patient characteristics, e.g. age and gender if known. Please try to avoid too many abbreviations in the text, tables and figures. Please make the tables and figures as self-explanatory as possible. * Forestplots: Please add number of patients (i.e. 2x2 table data) if available, add self-explanatory text on the x-axis and explain the direction - is a higher or lower odds ratio better or worse? * I suggest to either use a fixed-effect or random-effects model for all analysis. As I understand it, you selected the type of model based on heterogeneity statistics. Maybe it is better to use the same type of model in all analysis.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Comments to the Author:
The reviewer really appreciates having an opportunity to evaluate this manuscript entitled "Undifferentiated-predominant mixed-type early gastric cancer is more aggressive than pure undifferentiated-type: a meta-analysis" written by Yang Peng et al. The manuscript is generally interesting. But the reviewer has major and minor comments listed below.

Major comments:
1. The authors conducted a meta-analysis regarding lymph node metastasis, SM invasion, lympho-vascular invasion, and gross type compared between MU and PU. To the best of the reviewer's knowledge, the majority of the literature included in this study is single center, retrospective studies. In addition, due to the small number of references, the level of evidence obtained from this study seems to be low.
Answer: Thank you for pointing this out. We have updated our search results to November 2021, and two new papers were added through retrieval. There are only 12 articles in the current study, and we have included the literature on all relevant aspects to investigate this issue.

2. The information extracted from the literature is insufficient. Reference 18 describes depth, LVI, and ulcer findings. Reference 21 describes the depth (all cases SM) and the gross type. Reference 27 describes the depth (M in all cases). Reference 28 describes the gross type and ulcer findings. However, the authors did not indicate these data in Table 1 and did not include them in the analyses. Therefore, the authors seem to have arbitrarily selected the references for analysis. The information in each literature should be extracted again.

Answer: Thanks for reading this paper carefully. We have extracted the data from each piece of literature again. Reference 18 only describe the depth, LVI, and ulcer findings in intramucosal EGC. As for the Reference 21 and 27, we have separated the M cases and the SM cases in Table 1. Reference 28 only describes the ulcer findings in cases that are LVI(-).

3. The clinical benefits of this meta-analysis are unclear. Although this study was conducted on surgical cases, the clinical merits should be clearly shown, such as the reduction of lymph node dissection in PU or the necessity of D2 dissection in MU.
Answer: Thank you for your comments. We agree with you and have made corresponding modifications according to your comments.

Minor comments:
4. The introduction is redundant. If the authors would like to discuss surgical cases, the description regarding ESD is too long. The following part should be deleted. ESD has become an important method of treatment for EGC patients with a lower risk of LNM…which consists of both differentiated and undifferentiated histological type according to the histology.
Answer: We have removed the corresponding paragraph.

5. In the Methods section, why is the gross type included in the comparison? The authors should provide rationale that the gross type is involved in lymph node metastasis. In addition, the reason for dividing the patients into 0-IIc and others is unclear. The authors should clearly state why they divided the patients in such fashions.
Answer: At the beginning of the data extraction, we had wanted to compare PU and MU with respect to the gross type, thanks for your reminder. We read the 12 studies again and found that the gross type was not associated with lymph node metastasis in the multivariate analysis, and have therefore removed the relevant description.

6. In the Results section, why did the authors analyze the literature from Japan or China as sub-analyses should be presented.
Answer: Most of the studies are from China and Japan, and only these articles can be used for subgroup analysis. In addition, these subgroup analyses were performed to clarify the source of heterogeneity.

7. In the Discussion section which is redundant, the following sentence is also found in the Introduction section. The reviewer recommends that the authors delete it. ESD has been gaining popularity as the primary treatment for patients with EGC. However…. provide clear guidance for the clinical management of MU and PU EGC.
Answer: We have removed the corresponding paragraph.

8. The authors also describes “Another study reported that the protein expression levels of Ki-67, extracellular matrix metalloproteinase inducer (EMMPRIN) and vascular endothelial growth factor (VEGF), which participate in angiogenesis and cell proliferation, were reduced in intestinal-type GC compared to the intestinal portion of mixed-type GC [32]”. However, the mixed-type here means the gastro-intestinal type in mucus traits. It has nothing to do with this study. The author should delete the description.
Answer: We have removed the corresponding paragraph.

9. The conclusion is too abstract. The authors should clearly state what the results mean in clinical practice.
Answer: We have made the corresponding modifications in the conclusion.

Reviewer: 2
Comments to the Author:
In the present manuscript, Peng Yang et al. have compared the biological behavior of undifferentiated-predominant mixed-type (MU) EGC and pure undifferentiated-type (PU) EGC. This study identified that patients with MU EGC had an increased risk of submucosal invasion, lymphovascular invasion, depressed-type (IIc) endoscopic growth patterns and LNM compared with PU EGC patients.
Manuscript is well written which will provide clarity to the scientific reader community. I am recommending this manuscript for publication after revision.

1. In the subgroup analysis, I suggest to be performed based on all countries not just for one country.
Answer: Thanks for your comments, we agree with you. But only articles from China and Japan which have enough data can be used for subgroup analysis.

2. Data and figure regarding the ulcer findings were not presented. Authors should provide the results of this part.
Answer: Data and figure regarding the ulcer findings have been presented in Table 1 and Fig 7.

3. I strongly suggest analysis based on survival outcomes to compare the mortality rate between MU EGC and PU EGC.
Answer: Thank you for your comment. Unfortunately, relevant data were lacking when we extracted the information from these studies to do this comparison.

Reviewer: 3
Comments to the Author:
Statistical review:
Thank you for an interesting study. It should be of interest, but I have a few comments about the statistical analysis and presentation of the results.

1. Please also use funnelplots to assess publication bias. You may present some of them in the manuscript.
Answer: Thanks for your comment. We have presented one funnel plot (Figure 8) as part of the main results.

2. I suggest to add an example of the stata codes used for the statistical analysis. It makes the reproduction of your analysis easier.
Answer: We have added stata codes in the *Methods- Statistical analysis* section.

3. Table 1: I suggest to add some general patient characteristics, e.g. age and gender if known. Please try to avoid too many abbreviations in the text, tables and figures. Please make the tables and figures as self-explanatory as possible.
Answer: We have made the corresponding revision in the article.

4. Forestplots: Please add number of patients (i.e. 2x2 table data) if available, add self-explanatory text on the x-axis and explain the direction - is a higher or lower odds ratio better or worse?
Answer: We have made the corresponding revision to the Figures.

5. I suggest to either use a fixed-effect or random-effects model for all analysis. As I understand it, you selected the type of model based on heterogeneity statistics. Maybe it is better to use the same type of model in all analysis.
Answer: We agree with you and used a random-effects model for all analyses.

VERSION 2 – REVIEW

REVIEWER	Pripp, Are Hugo Oslo universitetssykehus Ulleval, Oslo Centre for Biostatistics & Epidemiology
REVIEW RETURNED	02-Dec-2021

GENERAL COMMENTS	Statistical review: The statistical analysis seems well performed and described. If possible, could the forest plots be make even clearer and self-explanatory. For example, could you state directly in the plot what is treatment and control? If I only look at the forest plot without reading the rest of text, it is not quite clear whether undifferentiated-predominant mixed type of pure undifferentiated-type is intervention or control in the meta-analysis. Besides that, I have no other concerns and recommend the manuscript for publication.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 3
Comments to the Author:
The statistical analysis seems well performed and described. If possible, could the forest plots be make even clearer and self-explanatory. For example, could you state directly in the plot what is treatment and control? If I only look at the forest plot without reading the rest of text, it is not quite clear whether undifferentiated-predominant mixed type of pure undifferentiated-type is intervention or control in the meta-analysis. Besides that, I have no other concerns and recommend the manuscript for publication.
Answer: We have made the corresponding revision to the Figures.